

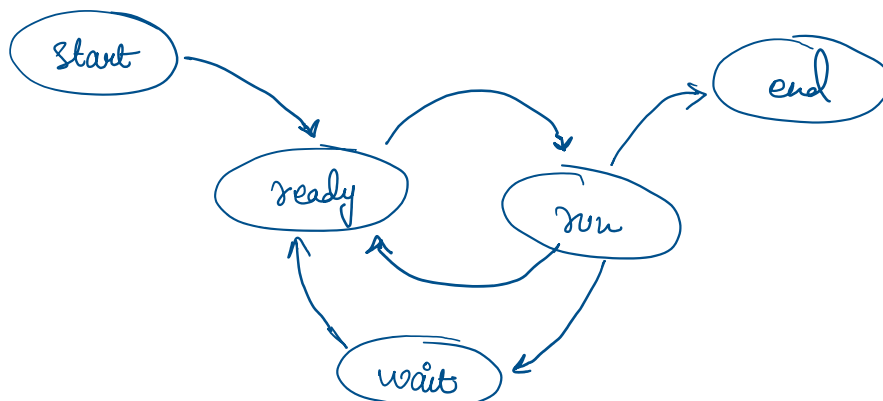
① What is a Process?

- A process is a program in execution.
- An entity which represents the basic unit of work to be implemented in the system.
- When a program is loaded into the memory, it becomes a process.
- Stack, Heap, Text, Data

② PCB

- Process control block is a data structure maintained by the operating system for every process.
- Maintained by Integer Process ID (PID)
- A PCB keeps all the info to keep track of a process also known as Task Control Block.
- Process ID
- Process State
- Process Privileges
- Pointer
- Program Counter
- CPU registers
- CPU scheduling information
- Memory Management information
- Accounting information
- I/O status information

③ Process States



(i) Start: The process is being created

(ii) Ready: The process is waiting to be assigned to a processor to be executed.

- ready : The process is waiting to be assigned to a processor to be executed.
- (iii) Running : The process is assigned to a processor by the OS scheduler, the process state is set to running & the processor executes its function.
- (iv) Waiting : The process is waiting for some event to occur.
- (v) Terminate : Once the process finishes its execution, it is moved to the terminated state, where it is waiting to be removed from the main memory.
- (vi) Suspend Ready : The process is removed from the main memory into a suspended state by the scheduler. The process will transition back to ready state when it is brought back to the main memory.
- (vii) Suspend Wait/Block : Similar to Suspend ready, but is queued to back to waiting stage. When the work is finished it may go to suspend ready.
- (viii) CPU or I/O Bound Process : When the work is intensive in terms of CPU or I/O ~~bound~~ operations.

④ CPU Scheduler

Job scheduler

- (i) Long-term Scheduler : decides how many programs are to be kept in ready state. Decides the degree of multi-programming
 - (ii) Short-term Scheduler : Will decide which process is to be executed next & then it will call the dispatcher. (ready-run & vice-versa)
(software)
 - (iii) Medium Scheduler : Suspension decision is made by the medium scheduler. Used for swapping process from main memory to secondary memory if vice-versa
- Process swapping sched

⑤ Context Switch

- A context switch is the mechanism to store and restore the state of a process in Process Control Block (PCB) so that a process execution can be resumed from the same point at a later time.
- Essential part of multi-tasking OS.
- When scheduler switches the CPU from executing one process to another, the state from the current running process is saved into PCB.
- Context switches are computationally intensive since register and memory space must be saved & restored.

→ Context switches are computationally intensive since register and memory space must be saved & restored.

⑥ Process Scheduling

→ the activity of process manager to select that handles the removal of the running process from the CPU and the selection of next process on the basis of a particular strategy

→ Scheduling Criteria - CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time.

→ Types - Preemptive & Non-preemptive

1. FCFS - First Come First Serve

- non-preemptive
- easy to understand & implement
- FIFO Queue
- Poor performance coz high average time

CRITERIA: Arrival Time

Q	PID	AT	BT	CT	TAT	WT (TAT - BT)
	P ₁	2	2	13	11	9
	P ₂	5	6	19	14	8
	P ₃	0	4	4	4	0
	P ₄	0	7	11	11	4
	P ₅	7	4	23	16	12

$$\begin{array}{c}
 P_3 \mid P_4 \mid P_1 \mid P_2 \mid P_5 \\
 0 \quad 4 \quad 11 \quad 13 \quad 19 \quad 23
 \end{array}$$

$$\text{Avg. TAT} = \frac{11 + 14 + 4 + 11 + 16}{5} = \frac{56}{5} = 11.2$$

$$\text{Avg. WT} = \frac{9 + 8 + 0 + 4 + 12}{5} = \frac{33}{5} = 6.6$$

2. SJF - Shortest Job First

- Easy to implement in Batch Systems
 - Impossible to implement in interactive systems
- ↳ preemptive & non-preemptive

BURST TIME

Q	PID	AT	BT	CT	TAT	WT
	P ₀	0	13	31	31	13
	P ₁	1	4	5	4	0
	P ₂	2	6	14	12	5
	P ₃	3	5	2	2	0

→ non-preemptive

35
49

$\begin{array}{c}
 P_0 \quad P_1 \quad P_2 \quad P_3 \quad P_3 \quad P_1 \quad P_1 \quad P_2 \quad P_2 \quad P_0 \\
 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad 14 \quad 31
 \end{array}$

$$\text{Avg WT} = \frac{13 + 0 + 5 + 0}{4} = \frac{18}{4} = 4.5$$

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$$\text{Avg TAT} = \frac{31+4+12+2}{4} = \frac{49}{4} = 12.25$$

3. SJF - preemptive

PID	BT	AT	CT	TAT	WT
P ₁	7	0	14	14	7
P ₂	3	1	4	3	0
P ₃	4	3	8	5	1

0 P₁ 1 P₂ 2 P₂ 3 P₃ 4 P₂ 5 P₃ 6 P₁ 14

4. Round Robin - preemptive

- time quantum
- Context switching

PID	AT	BT	CT	TAT	WT
P ₁	0	7	31	31	24
P ₂	1	4	9	8	4
P ₃	2	5	55	53	38
P ₄	3	4	56	53	42
P ₅	4	10	66	62	42
P ₆	4	4	50	46	37

Ready Queue P₁ P₂ P₃ P₄ P₅ P₆ P₁ P₃ P₄ P₅ P₆ P₂ P₄ P₅

Running Queue 0 P₁ 5 P₂ 9 P₃ 14 P₄ 19 P₅ 24 P₆ 29 P₁ 31 P₃ 36 P₄ 41 P₅ 46 P₆ 50 P₃ 55 P₄ 58 P₅ 62

5. Priority Based Scheduling